

ARUN KARTHICK P

Python | SQL | Machine Learning | Data Analysis | Data Preprocessing | Generative AI

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PROFESSIONAL SUMMARY

Computer Science Engineering student (3rd Year, CGPA 8.56) with hands-on experience in Artificial Intelligence, Python, SQL, Machine Learning, Data Analysis, Data Preprocessing, and Generative AI application development. Technical Manager at Data Forge, the official Data Science Club at SRMIST Trichy; 1st Place winner at the DND Data Arena 24-Hour Sprint. Skilled in data preparation, model development, testing, and implementation of Machine Learning solutions; performing Data Analysis and Data Preprocessing on large datasets; writing SQL and Python for data pipelines; and integrating Generative AI into real-world applications and dashboards.

TECHNICAL SKILLS

Python: Pandas, NumPy, Scikit-Learn, Object-Oriented Programming, Scripting, Automation

SQL: PostgreSQL, Query Optimization, Joins, Aggregations, Relational Database Design

Machine Learning: Predictive Modelling, Classification, Regression, Random Forest, Model Evaluation, CTGAN

Data Analysis: Exploratory Data Analysis, Trend Identification, Root Cause Analysis, Power BI, Tableau, DAX

Data Preprocessing: Data Cleaning, Data Validation, Data Transformation, ETL Pipelines, Power Query

Generative AI: AI API Integration, Prompt Engineering, LLM-Powered Insight Generation, AI-Driven Dashboards

Tools & Platforms: Streamlit, Plotly, DuckDB, Apache Parquet, GitHub, RapidMiner, Advanced Excel

PROJECT EXPERIENCE

SYNAPSE — Synthetic Data Intelligence Platform

Python, SQL, Machine Learning, Data Preprocessing | 2025

- Architected an end-to-end synthetic data pipeline (event to silver to gold) using SQL-based DuckDB queries and CTGAN-based Machine Learning generation, producing datasets up to 50,000+ rows in under 2 minutes.
- Applied Data Preprocessing and Data Analysis techniques to engineer a composite Trust Score combining distribution similarity, correlation fidelity, and ML utility validation, achieving scores of ~0.7.

TATA IPL EcoDot Analytics

Python, Machine Learning, Data Analysis | 2026

- Built a Python ETL pipeline for Data Preprocessing on 1,200+ raw match JSON files (290k+ delivery rows); handled model development, training, and testing for a Machine Learning Random Forest Regressor to forecast matchup-level environmental metrics.
- Performed Data Analysis to design a dynamic multiplier mapping system and a dark-themed Streamlit dashboard with a proprietary Green Impact Score ranking metric.

Digital Wellbeing AI Dashboard

Python, Generative AI, Data Analysis | 2025

- Built a Generative AI-powered Streamlit dashboard performing Data Analysis on digital wellbeing patterns from behavioural data, surfacing real-time insights on screen time and usage habits.

Freelancer Platform Funnel & Growth Paradox Analysis

SQL, Data Analysis, Data Preprocessing | 2025

- Performed Data Analysis and Data Preprocessing on 35,000+ interaction records using SQL and Power BI to diagnose revenue-efficiency vs. growth trends and funnel conversion leakage; built a cohort-driven ETL workflow.

WORK EXPERIENCE

Data & Analytics Intern — Right Angle Solutions

Jun 2026 – Jul 2026

- Validated and analysed financial and operational data for the Finance & Accounts team, preparing structured reports to support accurate stakeholder decision-making.
- Performed data quality checks on spreadsheet-based records to identify discrepancies and improve reporting accuracy.

Digitalization & Documentation Volunteer — Southern Railway, Coimbatore Jn.

Jun 2026 – Jul 2026

- Digitalized station records and developed a lightweight offline application for file categorization, improving record retrieval and scalability.
- Structured the application's categorization logic and offline-first architecture, enabling staff to sort and tag records without relying on internet connectivity.

LEADERSHIP

Technical Manager — Data Forge, Data Science Club, SRMIST Trichy

2024 – Present

- Lead technical workshops and project mentoring; drive end-to-end event management for data science and analytics competitions.
- Built and maintained the Data Forge club website, showcasing club events, projects, and technical initiatives.

ACHIEVEMENTS & CERTIFICATIONS

- 1st Place, DND Data Arena 24-Hour Sprint — Social Media Sentiment Track (Data n Dreads); solo entrant against team submissions.
- HackerRank SQL Gold Badge — Advanced SQL Certification • Microsoft DP-600 — Fabric Analytics Engineer Associate (907/1000)
- Google Data Analytics Professional Certificate • Microsoft ETL Fundamentals Certification
- Industry Simulations: Tata (Forage) GenAI-Powered Data Analytics • Deloitte (Forage) Data Analytics • EduSkills Google AI-ML Virtual Internship

EDUCATION

B.Tech Computer Science & Engineering — SRM Institute of Science & Technology, Tiruchirappalli

CGPA: 8.56 | 2024 – 2028